

HETTINGER, ND, USA

WMO: 727584

Lat: 46.014N

Lon: 102.655W

Elev: 825

StdP: 91.80

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 94038

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.3	-23.0	-29.4	0.3	-25.7	-26.5	0.4	-22.9	15.4	0.7	14.0	-1.8	3.3	300	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.7	34.6	19.5	32.5	19.1	30.6	18.6	22.1	29.9	21.0	28.6	20.0	27.6	5.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.0	26.1	18.5	14.7	24.6	17.4	13.8	23.3	68.9	30.0	64.5	28.5	60.7	27.5	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.1	10.8	-32.5	38.2	3.9	2.7	-35.3	40.2	-37.6	41.7	-39.7	43.2	-42.5	45.2		
(5)			-32.2	24.3	3.1	1.7	-34.4	25.5	-36.2	26.5	-37.9	27.5	-40.1	28.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-7.9	-7.9	-0.9	5.5	11.2	16.8	21.5	20.0	14.7	6.5	-0.4	-7.1
	DBStd	11.96	7.72	7.84	7.31	5.53	4.73	3.97	3.59	3.86	4.96	5.41	6.48	7.31
	HDD10.0	2582	553	500	340	152	42	2	0	13	134	314	531	
	HDD18.3	4715	812	733	596	386	226	73	11	28	131	368	563	789
	CDD10.0	1150	0	0	3	17	79	207	357	309	153	24	2	0
	CDD18.3	242	0	0	0	1	4	28	109	79	21	1	0	0
	CDH23.3	3738	0	0	2	20	98	456	1501	1168	451	39	2	0
	CDH26.7	1602	0	0	0	4	21	151	704	530	183	9	0	0
(14) Wind	WSAvg	4.6	4.8	4.4	4.9	5.3	5.3	4.6	4.1	4.1	4.3	4.6	4.4	4.6
(15) Precipitation	PrecAvg	402	8	7	14	41	70	87	54	38	35	28	11	8
	PrecMax	686	26	37	54	134	208	202	124	94	156	126	40	23
	PrecMin	209	0	0	3	0	4	19	3	2	2	0	0	0
	PrecStd	104	6	7	11	31	45	44	32	28	35	32	10	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	14.0	22.2	26.2	29.0	33.9	38.7	36.9	34.7	27.6	21.2	11.7
		MCWB	5.3	6.8	10.1	12.9	17.2	20.6	20.3	19.8	18.0	14.4	10.4	5.2
	2%	DB	6.7	8.4	17.6	22.1	26.2	30.7	35.2	34.0	31.2	22.9	16.3	7.6
		MCWB	2.7	3.3	8.4	11.6	16.0	19.1	19.8	19.2	16.9	12.2	7.9	2.8
	5%	DB	3.9	5.0	13.8	18.8	23.4	27.8	32.8	32.1	28.6	19.1	12.6	4.9
		MCWB	1.1	1.8	6.7	9.8	14.4	18.3	19.7	19.0	16.0	10.5	6.0	1.3
	10%	DB	2.4	2.6	10.0	16.0	21.1	25.9	30.8	29.8	25.6	16.2	9.4	2.5
		MCWB	0.1	0.2	4.7	8.3	13.4	17.2	19.3	18.3	15.2	9.2	4.1	0.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	7.1	11.2	14.0	19.0	23.0	23.9	22.8	20.6	15.4	10.4	5.5
		MCDB	10.8	13.5	19.9	23.8	26.2	29.9	32.1	32.2	29.8	24.9	20.2	10.7
	2%	WB	2.9	3.9	8.9	11.8	17.0	21.0	22.5	21.2	18.8	13.1	8.4	3.2
		MCDB	6.2	7.6	16.7	20.2	23.7	27.5	30.3	29.7	27.4	20.6	15.8	7.2
	5%	WB	1.3	1.9	6.9	10.4	15.3	19.6	21.4	20.1	17.4	11.4	6.3	1.5
		MCDB	3.7	4.8	13.6	17.7	22.0	26.0	29.0	28.3	25.5	18.0	12.3	4.7
	10%	WB	0.2	0.4	4.9	8.9	13.9	18.2	20.5	19.2	15.8	9.7	4.4	0.1
		MCDB	2.3	2.6	9.9	15.1	19.9	24.4	28.0	27.2	23.6	15.8	8.9	2.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.7	12.7	14.1	13.9	14.2	15.7	16.4	16.2	14.2	13.6	11.4
		MCDBR	13.2	14.7	19.3	19.7	18.0	17.6	19.1	19.6	21.3	20.4	19.1	14.2
		MCWBR	9.8	10.3	11.2	10.9	9.5	8.4	7.2	7.9	9.0	11.0	11.6	10.2
	5% WB	MCDBR	12.6	13.8	18.5	18.1	15.8	15.3	15.9	16.8	18.1	18.5	18.8	13.7
		MCWBR	9.7	10.0	11.1	10.3	8.8	8.2	7.5	7.6	8.3	10.1	11.6	10.1
(40) Clear-Sky Solar Irradiance	taub		0.253	0.259	0.278	0.316	0.338	0.349	0.361	0.366	0.326	0.296	0.264	0.248
	taud		2.521	2.530	2.530	2.425	2.430	2.433	2.434	2.390	2.485	2.543	2.529	2.528
	Ebn at Noon		880	943	963	944	928	916	899	881	894	881	856	849
	Edh at Noon		62	75	88	107	111	111	110	110	92	74	62	56
(44) All-Sky Solar Radiation	RadAvg		1.74	2.74	3.85	4.97	5.86	6.70	6.99	5.84	4.48	2.89	1.87	1.39
	RadStd		0.12	0.23	0.31	0.29	0.57	0.50	0.28	0.35	0.30	0.30	0.20	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page